

Meeting Scheduling Platform

Requirement Analysis & Core Features

1 Product Vision

Build the first tool that genuinely unifies 1-on-1 booking automation (Calendly, Cal.com) and group availability polling (Doodle, When2meet) in one product — closing the manual handoff that exists in every competitor studied, with a responsive, timezone-explicit foundation.

Target users: Individuals/freelancers and teams/organizations, served equally from day one. The platform must feel as fast as Calendly for a 1-on-1 link and as frictionless as When2meet for a group poll — without forcing either audience into a workflow built for the other.

2 Requirement Analysis

2.1 Problem Statement

Across the four tools studied, no single product handles both 1-on-1 booking and group polling well, none closes the loop from poll to confirmed calendar invite automatically, and timezone handling is inconsistent everywhere. This creates real, observed friction: teams use two tools and switch context, organizers manually re-key poll results into calendar invites, and participants get times displayed with no explicit timezone reference.

2.2 Stakeholders

- **Organizer (individual):** freelancer/professional sharing a personal booking link for 1-on-1 meetings.
- **Organizer (team lead):** polls a group, coordinates across time zones, needs the winning slot auto-confirmed.
- **Participant:** receives a link, has no account, needs to respond in under a minute.
- **Admin (org/team plan):** manages team members, branding, and integration settings.
- **Developer/integrator:** consumes the API/webhooks to embed scheduling into another product.

2.3 Functional Requirements

(Full detail on each item is in Section 3, Core Features — this list is the high-level summary.)

- The system shall let an organizer create both 1-on-1 booking pages and group polls.
- The system shall let participants vote on a poll without an account, via an open link (informal polls) or a unique named link per participant (required whenever the poll has a Required participant, so votes are reliably attributed).
- The system shall show a live, real-time heatmap of aggregated availability as votes come in.
- The system shall display every time in both the organizer's and the viewer's local timezone, explicitly labeled.
- The system shall auto-confirm a meeting and issue calendar invites once a slot has full agreement — instantly for 1-on-1 bookings, and for polls once every Required participant has voted Yes.
- The system shall let the organizer set a custom deadline per poll, and shall notify the organizer (not auto-decide) if a Required participant hasn't voted Yes by then, so slots can be revised and all participants informed before re-voting.
- The system shall send automated reminders across the poll/booking lifecycle.

- The system shall support a shared team dashboard, round-robin scheduling, and mutual calendar checks for two-party bookings.
- The system shall support calendar sync (Google/Outlook/Apple), video link generation, and a documented API with webhooks for CRM/automation integrations.

2.4 Non-Functional Requirements

- **Performance:** real-time heatmap updates should reflect a new vote within ~ 1 second.
- **Reliability:** calendar sync and auto-booking must be idempotent — a webhook retry or duplicate event must not create duplicate calendar invites.
- **Privacy:** participant data collection kept minimal (name + optional email only) to preserve the low-friction, no-account experience competitors are praised for.
- **Security/Compliance:** SOC2-ready data handling practices from day one; formal SOC2/HIPAA certification targeted for a later org-tier milestone.
- **Accessibility:** tap/click targets and the heatmap's color coding must meet WCAG AA contrast and touch-target size guidelines on all screen sizes.
- **Scalability:** real-time layer must handle poll-heavy usage spikes (e.g., a 50-person poll voting within minutes) without degraded update latency.

2.5 Constraints & Assumptions

- Web-only — no native mobile app; all interaction (including availability selection) must work as a responsive, tap/click-friendly web experience.
- No self-hosting or on-prem distribution.
- No payment collection.
- Free tier is uncapped on event types/polls and ad-free.

3 Core Features

3.1 Event Creation

- Create a 1-on-1 booking page: title, duration, description, available hours, buffer time, location/video link.
- Create a group poll: title, description, a set of proposed date/time slots (specific dates or recurring weekly pattern).
- Both flows share the same underlying “event” object so a poll can be converted into a recurring booking page later.
- Support unlimited event types and unlimited polls on the free tier, with no ads.

3.2 Timeslot Selection & Voting

- Click/tap availability grid for participants, responsive across desktop and mobile browsers.
- Voting options for group polls: Yes / No / If-need-be, matching Doodle's nuance rather than a binary yes/no.
- Live, color-coded heatmap of overlapping availability, updated in real time as participants respond.
- No account required for participants to vote or book. Identification instead comes from the link: an open link for informal polls (self-entered name), or a unique named link per invited participant whenever the poll has a Required participant — this is how the system reliably knows which vote belongs to whom without a login.

- 15-minute interval granularity for slot selection.

3.3 Timezone Handling (Explicit, First-Class)

- Every event stores its timezone; every participant sees times converted to their detected local timezone automatically.
- Organizer and participant local times are shown side-by-side wherever a time is displayed — never assumed to be shared.

3.4 Auto-Confirmation & Booking

- When a poll closes (manually or by deadline), the winning slot is automatically converted into a confirmed meeting.
- Calendar invites are generated and sent to all participants automatically — no manual “send invite” step.
- 1-on-1 bookings confirm and sync instantly on selection, matching Calendly/Cal.com’s flow.

3.5 Notifications & Reminders

- Automated email reminders for: poll invitation, poll closing soon, poll result/confirmation, upcoming meeting.
- Optional SMS reminders (stretch goal, tiered).

3.6 Team & Org Features

- Shared team dashboard showing both individual booking pages and group polls in one place.
- Round-robin and collective scheduling for team booking pages.
- Basic admin panel: manage members, view usage.

3.7 Mutual Calendar Booking (Two-Party)

- When an organizer invites a specific, named person (not an open public link), the system checks both parties’ connected calendars for free/busy overlap and only surfaces mutually free slots.
- The invitee sees only available windows, never the organizer’s full calendar detail.
- Booking confirms and syncs to both calendars instantly on selection, same as standard 1-on-1 booking.

3.8 Required Participants & Conflict Resolution (Polls)

- Organizer can flag one or more poll participants as Required (e.g., a client) versus optional/flexible (e.g., internal teammates).
- Organizer sets a custom response deadline for each poll.
- Auto-confirmation requires every Required participant to vote Yes on the slot — an “If-need-be” vote from a Required participant does not count as agreement. For optional participants, “If-need-be” is treated as available and does not block confirmation.
- If a Required participant hasn’t voted Yes on any slot by the deadline (No, If-need-be only, or no response), the poll does not auto-finalize: the organizer is notified, can change or add candidate slots, and all participants are informed of the update before the poll re-opens for voting.

3.9 Integrations (Calendar + Video + CRM/Webhooks)

- **Calendar sync:** two-way sync with Google Calendar, Outlook/Microsoft Graph, and Apple Calendar (CalDAV).

- **Video conferencing:** auto-generated links for Zoom, Google Meet, and Microsoft Teams on confirmed bookings.
- **CRM/automation:** outbound webhooks on key events (poll closed, booking confirmed, booking canceled) plus a documented REST API, enabling Zapier-style automation and CRM integrations (Salesforce/HubSpot-class) without building bespoke connectors for each integration.
- **Embeddable widget:** booking page and poll can be embedded in an external website/app.

3.10 Branding & Access

- No ads on any tier, including free.
- Custom booking page URL slug for every user (not gated to a paid custom-domain tier).
- White-label options planned for team/org tiers.